Sahaj Bhadja

(213) 284-4651 bhadja@usc.edu sahaj-bhadja sahaj-bhadja sahaj-bhadja

Summary

Software engineer with 2+ years of experience building and sustaining **distributed systems software** across **C/C++, Python, SQL, and Linux-based environments**, with strong focus on multithreaded workflows, debugging, testing, and high-quality release delivery for mission-critical platforms.

Education

University of Southern California **Los Angeles, CA**
Master of Science in Computer Science *August 2024 – May 2026*
GPA: 3.75/4 Coursework: Analysis of Algorithms, Database Systems, Web Technologies

PDEU
B.Tech. in Computer Science and Engineering *Nov 2020 – May 2024*
GPA: 3.88/4 Coursework: Object-Oriented Programming, Software Engineering, Operating Systems, Data Retrieval

Experience

USC Dornsife **LA, California**
Software Engineer *September 2025 – Present*
- Built and sustained distributed backend services across **FastAPI**, **Azure SQL**, and **PostgreSQL** for a multi-tenant platform supporting 500+ users, reducing dashboard latency from 1.2s to 350ms through schema and query optimization.
- Engineered multithreaded background-job and integration workflows for 5K+ monthly tasks with validation and escalation-friendly diagnostics, improving throughput 3x while sustaining 99%+ production reliability.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*
- Developed cross-platform microservice software with 20+ unit/integration suites using **Jenkins**, **PyTest**, and Mockito across Linux/Windows/macOS, improving repeatability and regression coverage in release cycles.
- Troubleshot API and ETL escalations using logs, reproducible scenarios, and OOP design patterns across **AWS Glue/S3** workflows, improving response times by 30% and pipeline efficiency by 68%.

Grad8 **LA, California**
Software Engineer Intern *June 2025 – August 2025*
- Implemented concurrent processing and robust API workflows for an AI platform using **Flask**, secure ingestion, and **AWS RDS**, handling 10K+ daily requests under high-load conditions.
- Contributed to definition-through-release delivery with CI/CD-backed validation, lowering feedback turnaround by 60% while maintaining stable performance and early-support readiness.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 – May 2025*
- Built Python-based system pipelines for autonomous planning, processing 100K+ trajectory logs and improving long-horizon performance by 18% through iterative test-debug cycles.
- Designed spatial-temporal evaluation workflows and documented reproducible diagnostics, improving closed-loop stability by 15% and enabling faster issue isolation across team collaborations.

LST Capitals **India**
Web Developer Intern *November 2022 – January 2023*
- Built backend APIs and distributed synchronization flows for a booking/payments platform across 50K+ records, covering reservation, pricing, and state-management scenarios under production constraints.
- Mapped 12+ integration points and collaborated with peer teams on deployment scenarios, improving reproducibility, stakeholder alignment, and release confidence.

Skills

Languages: C/C++, Python, Java, JavaScript/TypeScript, SQL
Systems Programming: Unix/Linux System Programming, Multithreaded Programming, OS Internals, Concurrency Patterns
Backend & APIs: FastAPI, Flask, Node.js, REST APIs, JSON, OOP, Microservices, Escalation Triage
Databases & Storage: PostgreSQL, MySQL, MongoDB, SQL Server, Azure SQL, Relational Data Modeling, Storage-Domain Concepts
Cloud & Infrastructure: AWS (RDS, S3, Glue), Microsoft Azure, Docker, Kubernetes, Jenkins, Git/GitHub
Quality & Delivery: Unit Testing, Integration Testing, Regression Coverage, CI/CD, Debugging, Early Customer Support, Sound Engineering Practices
Engineering Foundations: Data Structures, Algorithms, Complexity Analysis, Cross-Functional Collaboration, Deadline-Oriented Execution

Projects

CPU-Scheduling Simulator | *C++, React.js, JavaScript*
- Implemented and validated 6 scheduling algorithms with 1K+ simulated process combinations, reducing debugging time by 20% while strengthening operating-system and multithreading reasoning.

Chat2SQL | *Java, TypeScript, SQL, Docker, Maven*
- Built and tested an AI-assisted data workspace across Java + TypeScript services supporting 10+ databases, with structured query execution and reproducible validation across environments.

Big-Brain | *LLaMA-2, FAISS, PostgreSQL, GCP*
- Engineered distributed retrieval/indexing pipelines over 50K+ documents with sub-second search performance, improving retrieval latency by 40% through storage and indexing optimization.